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General Knowledge

World Geography

STUDY NOTES

Climatology- Pressure and Winds

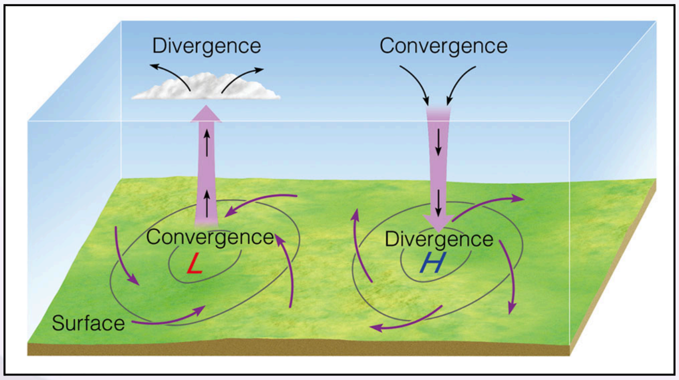
CLIMATOLOGY- PRESSURE AND WINDS

Pressure and Wind

- The velocity and direction of the wind are the net results of the wind-generating forces.
- The winds in the upper atmosphere, 2 to 3 km above the surface, are free from the frictional effect of the surface and are controlled mainly by the pressure gradient and the Coriolis force.
- When isobars are straight and when there is no friction, the pressure gradient force is balanced by the Coriolis force and the resultant wind blows parallel to the isobar. This wind is known as the **geostrophic wind**.
- The **wind circulation around a low** is called **cyclonic circulation**.
- **Around a high,** it is called **anti-cyclonic circulation**.
- The direction of winds around such systems changes according to their location in different hemispheres.

Pressure System	*Pressure Condition at the Centre*	*Pattern of Wind Direction*	
		Northern Hemisphere	*Southern Hemisphere*
Cyclone	Low	Anticlockwise	Clockwise
Anticyclone	High	Clockwise	Anticlockwise

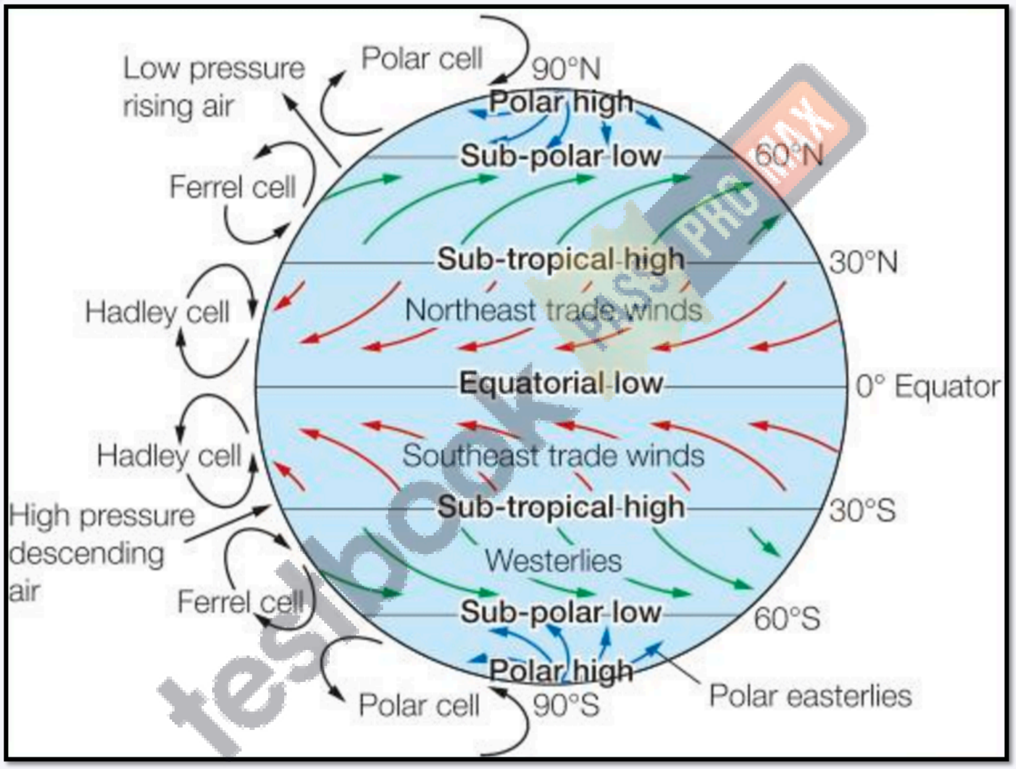
- The wind circulation at the earth's surface around low and high on many occasions is closely related to the wind circulation at higher levels.
- Generally, **over a low-pressure area, the air will converge and rise**. Over **high-pressure areas, the air will subside from above and diverge** at the surface.



- Apart from convergence, some eddies, convection currents, orographic uplift, and uplift along fronts cause the rising of air, which is essential for the formation of clouds and precipitation.

General Circulation of the Atmosphere:

The pattern of the movement of the planetary winds is called the general circulation of the atmosphere. The general circulation of the atmosphere also sets in motion the ocean water circulation which influences the earth's climate.



The wind movement in the atmosphere may be classified into three broad categories:

Primary circulation:

- It includes **planetary wind** systems which are related to the general arrangement of pressure belts on the earth's surface. In fact, it is the primary circulation patterns that prepare the broad framework for the other circulation patterns.

Secondary circulation:

- It consists of **cyclones, anti-cyclones,** and **monsoons**.

Tertiary circulation:

- It includes all the **local winds** which are produced by local causes such as topographical features, the influence of the sea, etc. Their impact is restricted to a particular area.

Planetary Winds

Planetary winds are large-scale winds that blow in the same direction all year round. These winds are controlled by **pressure belts** and are divided into three main types: **Trade Winds**, **Westerlies**, and **Polar Easterlies**. Here are the key features of these winds:

- **Planetary winds** blow constantly in one direction year-round.
- They are also called **prevailing winds** because they are steady and consistent.
- These winds always flow from **high-pressure** areas to **low-pressure** areas over the Earth's surface and oceans.

Types of Planetary Winds

1. Trade Winds (Tropical Easterlies)

The **trade winds** blow from **high-pressure areas** near 30° latitude (both north and south of the equator) toward the **equatorial low-pressure belt**. These winds are found between **30°N and 30°S** latitudes.

- In the **Northern Hemisphere**, they are called **northeast trade winds**, and in the **Southern Hemisphere**, they are called **southeast trade winds**.
- The **Coriolis effect** causes the winds to curve, so they don't blow directly north-south.
- **Trade winds** are warm and carry moisture. As they travel toward the equator, they get warmer and more humid.
- When the trade winds from both hemispheres meet at the **equator**, they **converge** and rise, causing **heavy rainfall**.

Key points:

- Blow from 30°N/S toward the equator.

- Meet at the equator and cause rain.
- Humid and warm winds.

2. Westerlies

The **westerlies** are winds that blow from the **subtropical high-pressure zones** toward the **subpolar low-pressure zones**, found between **30° and 60° latitude**.

- In the **Northern Hemisphere**, they blow from **southwest to northeast** and in the **Southern Hemisphere**, from **northwest to southeast** due to the **Coriolis effect**.
- These winds bring **rain**, especially to the **western coasts** of continents.
- The **Southern Hemisphere's westerlies** are stronger and more consistent than those in the Northern Hemisphere, as there are fewer landmasses to block their path.

Key points:

- Blow from subtropical high to subpolar low-pressure areas.
- Bring rain to the western coasts of continents.
- Stronger and more consistent in the Southern Hemisphere.
- Known as **Roaring Forties**, **Furious Fifties**, and **Shrieking Sixties** because of strong storms.

3. Polar Easterlies

Polar easterlies are cold, dry winds that blow from the **high-pressure areas** near the poles toward the **low-pressure areas** at the subpolar regions.

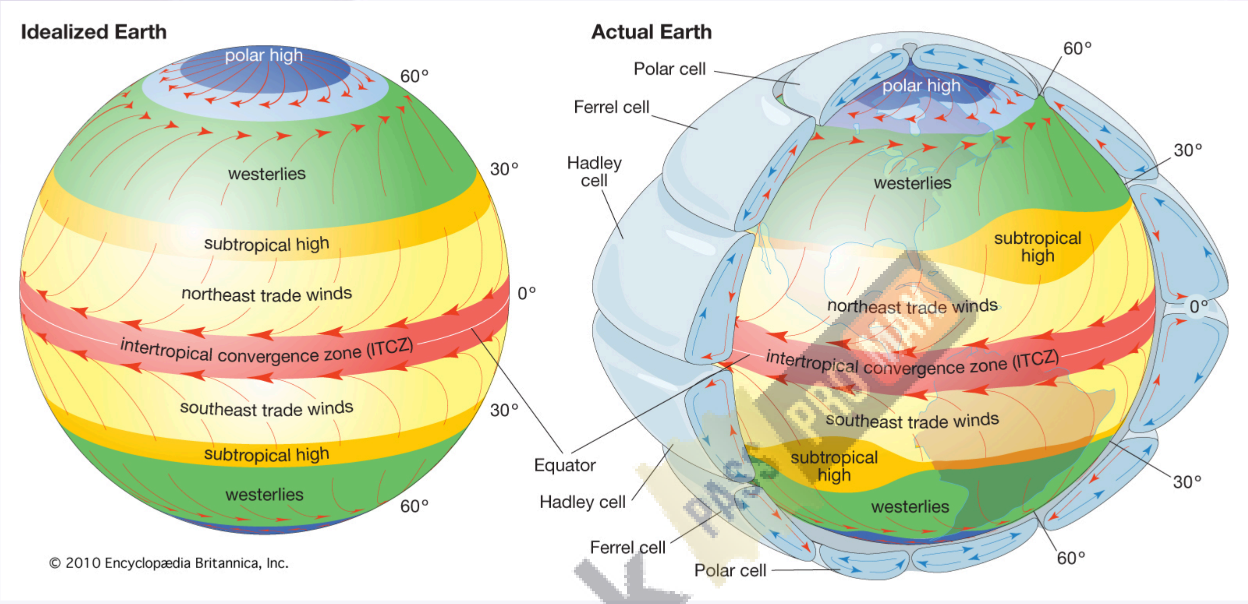
- In the **Northern Hemisphere**, these winds blow from the **northeast**, and in the **Southern Hemisphere**, they blow from the **southeast**.
- These winds are **very cold** and dry, and they don't bring much rain. However, when they meet the **westerlies**, they can create **cyclones** (storms).
- Polar easterlies cause **rapid weather changes** and can lead to heavy **rainfall** when they mix with westerlies.

Key points:

- Blow from the poles to the subpolar regions.
- Very cold and dry.

- Cause sudden weather changes and can bring rain when they meet westerlies.

Pattern of Planetary Winds



- Primary or planetary winds blow **from high-pressure belts to low-pressure belts** in the same direction throughout the year. They blow over a vast area of continents and oceans.
- **Trade winds, westerlies, and polar easterlies** together form the planetary wind circulation.
- The air at the **Inter Tropical Convergence Zone (ITCZ)** rises because of convection caused by **high insolation** and a **low pressure** is created.
- **The winds from the tropics converge at this low-pressure zone.** The converged air rises up to an altitude of 14 km and moves towards the poles. This causes accumulation of air at about 30°N and S.
- Part of the accumulated air sinks to the ground and forms a **subtropical high**. Another reason for sinking is the cooling of air when it reaches 30°N and S latitudes.
- Near the land surface, the air flows toward the equator as the easterlies or tropical easterlies, or trade winds. Because of the Coriolis force, their direction becomes **north-east and south-east in the northern and southern hemispheres respectively**.
- The easterlies from either side of the equator converge in the Inter Tropical Convergence Zone (ITCZ). Thus, winds originating at ITCZ come back in a circular fashion. Such a cell in the tropics is called the **Hadley Cell**.

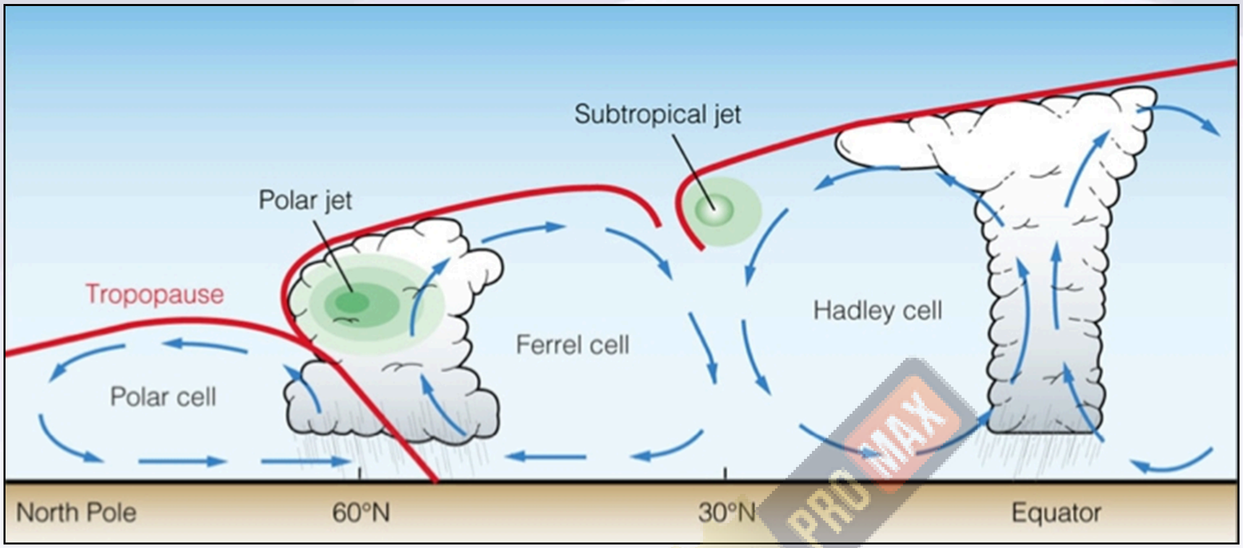
- In the middle latitudes (30° to 60°) the circulation is that of sinking cold air that comes from the poles and the rising warm air that blows from the subtropical high pressure belt.
- These winds are deflected due to Coriolis force and become **westerly** in both hemispheres. **The deflected wind is called westerlies.**
- These winds meet along the sub-polar low-pressure belt and rise high in the troposphere. From here, air moves away in both directions – towards the pole and the equator.
- These winds start descending down above the sub-tropical high-pressure belt and polar high-pressure belt to form cells. These cells are called **Ferrel cells** and **Polar cells** respectively.
- The prevailing westerlies are relatively more variable than the trade winds both in direction and intensity.
- There are more frequent invasions of polar air masses along with the traveling cyclones and anticyclones. These moving cells of low and high pressures largely affect the movement of westerlies.
- The **westerlies are stronger in the colder regions**. In the southern hemisphere, westerlies are so powerful and persistent due to the absence of land between **40° - 60°S** that these are called '**roaring forties**', '**furious fifties**', and '**screaming sixties**' along 40°S, 50°S and 60°S latitudes.
- Winds move away from polar high pressure to sub-polar low pressure along the surface of the earth in the polar cell. Their direction becomes easterlies due to Coriolis force. These are called **polar easterlies**.
- Winds coming from the subtropical and the polar high belts converge to produce cyclonic storms or low-pressure conditions. This zone of convergence is also known as the **polar front**.

The pattern of planetary winds largely depends on:

- Latitudinal variation of atmospheric heating
- Emergence of pressure belts
- The migration of belts following the apparent path of the sun
- The distribution of continents and oceans
- The rotation of the earth.

Circulation

Hadley Cell, Ferrel Cell



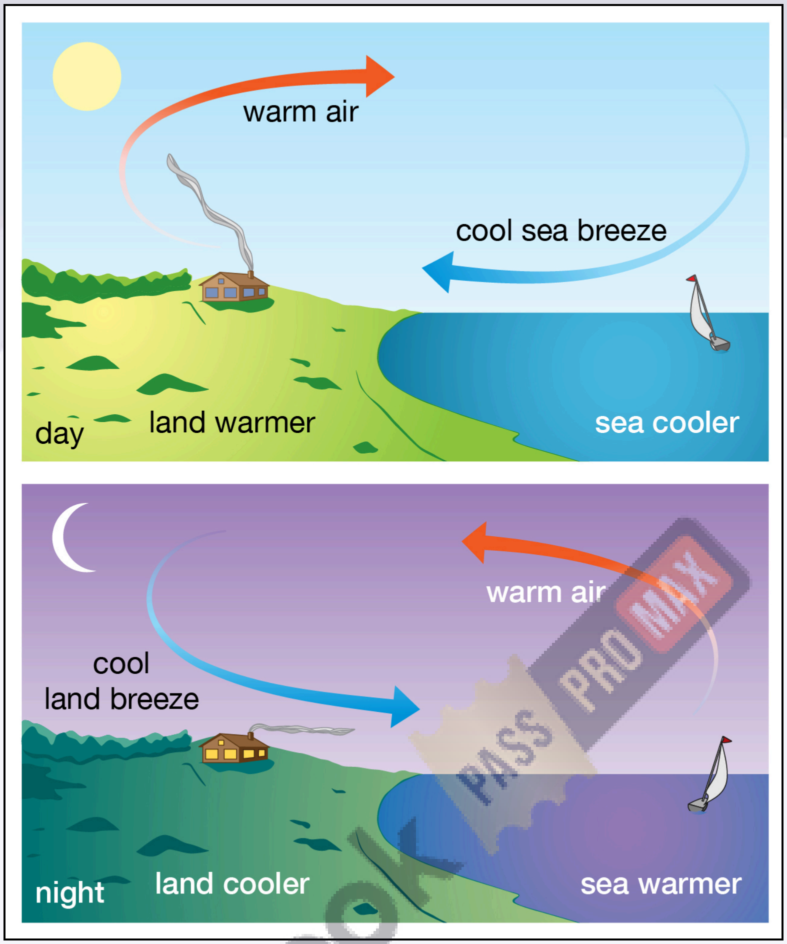
Seasonal Winds

- The pattern of wind circulation is modified in different seasons due to the shifting of regions of maximum heating, pressure, and wind belts.
- The most pronounced effect of such a shift is noticed in the **monsoons**, especially over southeast Asia....

Local Winds

- The winds which are produced by purely local factors are called local winds.
- These local winds play a significant role in the weather and climate of a particular locality.

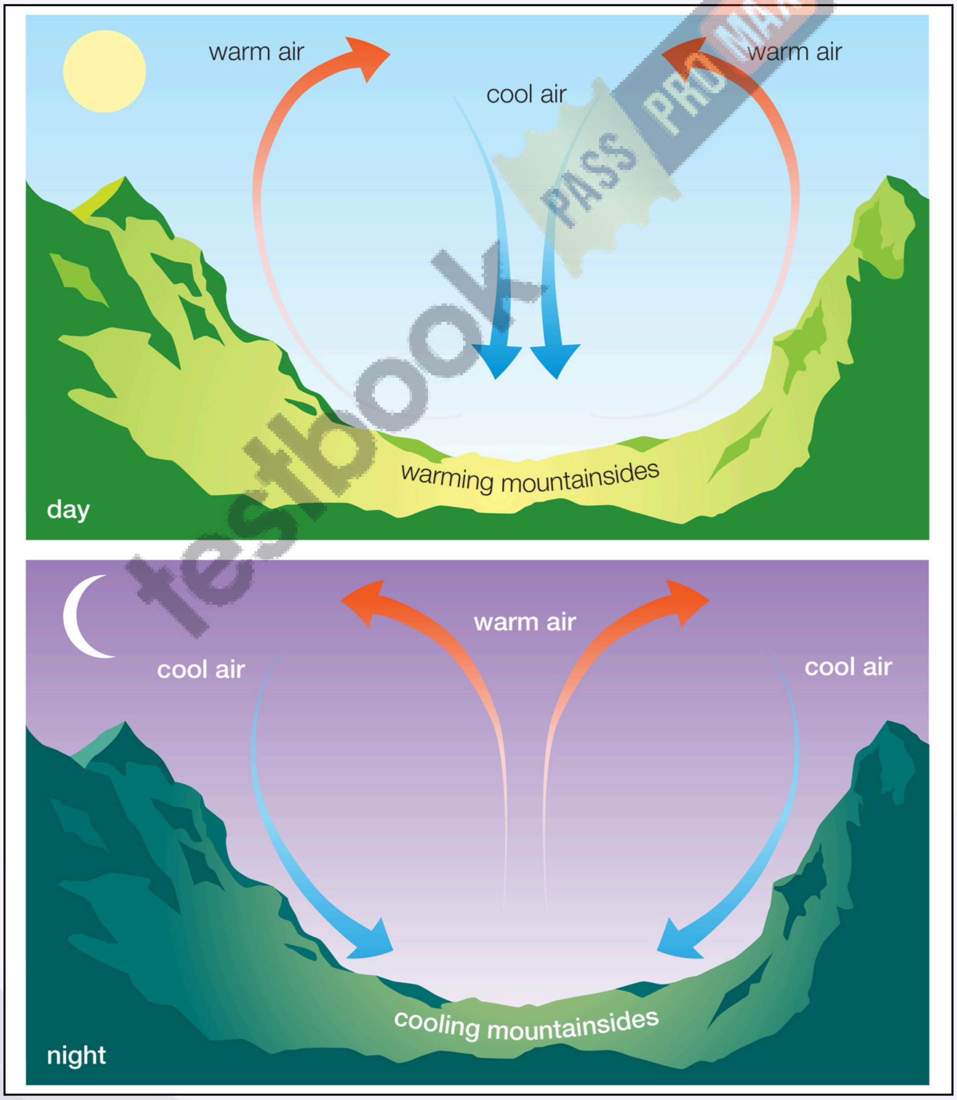
Land and Sea Breezes



- The land and sea absorb and transfer heat differently.
- During the **day the land heats up faster and becomes warmer than the sea**. Therefore, over land, the air rises giving rise to a low-pressure area, whereas the sea is relatively cool, and the pressure over the sea is relatively high.
- Thus, a **pressure gradient from sea to land** is created and the wind blows from the sea to the land as the sea breeze.
- **In the night the reversal of this condition takes place. The land loses heat faster and is cooler than the sea.**
- The **pressure gradient is from the land to the sea** and hence land breeze results.
- The land and sea breeze system is very shallow with an average depth of 1 - 2km.
- Over lakes, the height of circulation is much less.
- Warm tropical areas, where intense solar heating persists throughout the year, experience stronger and regular breezes compared to higher latitudes.

Mountain and Valley winds

- In mountainous regions, during the day the slopes get heated up and air moves upslope and to fill the resulting gap, air from the valley blows up the valley. This wind is known as the **valley breeze**.
- During the night the slopes get cooled and the dense air descends into the valley as the mountain wind.
- The cool air of the high plateaus and ice fields draining into the valley is called **katabatic wind**.
- Another type of warm wind occurs on the **leeward side of the mountain ranges**. The moisture in these winds, while crossing the mountain ranges condense and precipitate. When it descends down the leeward side of the slope the dry air gets warmed up by **adiabatic process**. This dry air may melt the snow in a short time.



Hot Local Winds

Local winds that are hot are caused by the advection of hot air from a warm source region. They may also be produced by the dynamic heating of air as it descends from an elevated area to a lowland.

A few well-known hot winds are:

Loo:

It is a hot and dry wind, which blows very strongly over the northern plains of **India and Pakistan** in the months of **May and June**. Their direction is from west to east and they are usually experienced in the afternoons. Their temperature varies between 45°C to 50°C. ‘

Foehn:

It is a strong, dusty, dry, and warm local wind that develops on the **leeward side of the Alps mountain ranges**. Regional pressure gradient forces the air to ascend and cross the barrier. Ascending air sometimes causes precipitation on the windward side of the mountains. After crossing the mountain crest, the Foehn winds start descending on the leeward side or northern slopes of the mountain as warm and dry winds. The temperature of the winds varies from 15°C to 20°C which helps in melting snow, thus making the pasture land ready for animal grazing and helping the grapes ripe early.

Chinook:

It is the name of a hot and dry local wind, which moves down the **eastern slopes of the Rockies** in the U.S.A. and Canada. The literal meaning of chinook is ‘**snow eater**’ as they help in melting the snow earlier. They keep the grasslands clear of snow. Hence, they are very helpful to ranchers.

Sirocco:

It is a hot, dry dusty wind, which originates in the **Sahara desert**. It is most frequent in spring and normally lasts for only a few days. After crossing the Mediterranean Sea, the Sirocco is slightly cooled by the moisture from the sea. Still, it is harmful to vegetation, and crops in that region. Its other local names are Leveche in Spain, Khamsin in Egypt, and Gharbi in the Aegean Sea area.

Harmattan:

It is a strong dry wind that blows over **northwest Africa** from the northeast. Blowing directly from the Sahara desert, it is a hot, dry and dusty wind. It provides a welcome relief from the moist heat and is beneficial to the health of people and hence known as '**the doctor**'. It is full of fine desert dust which makes the atmosphere hazy and causes problems to the caravan traders. It may cause severe damage to the crops.

Cold Local Winds

There are certain local winds that originate in the snow-capped mountains during winter and move down the slopes towards the valleys. A few of these are:

Mistral:

It originates in the **Alps** and moves over France towards the Mediterranean Sea through the Rhone valley. They are very cold, dry, and high-velocity winds. They bring down the temperature below freezing point in areas of their influence. As a protective measure, many of the houses and orchards of the Rhone Valley have thick rows of trees and hedges planted to shield them from the Mistral.

Bora:

It is a cold, dry north-easterly wind blowing down from the mountains in the **Adriatic Sea region**. It is formed due to the pressure difference between continental Europe and the Mediterranean Sea.

Blizzard:

It is a violent and extremely cold wind laden with dry snow. Such blizzards are a common occurrence in the **Antarctic**. Wind velocity sometimes reaches 160 kmph and temperature is as low as -70°C.

Details Local Winds (World)

Name	Region	Description
Alizé	West Central Africa	Maritime wet, fresh northerly wind
Bora	Eastern Europe to northeastern	Northeasterly wind

	Italy	
Cape Doctor	South African coast	The dry south-easterly wind that blows in summer
Chinook	Rocky Mountains	Warm, dry westerly wind
Diablo	San Francisco Bay, USA	Hot, dry, offshore wind from the northeast
Etesian	Greece and Turkey	Northerly wind
Elephanta	Malabar Coast, India	Strong southerly or southeasterly wind
Föhn	Alps and Northern Italy	Warm, dry southerly wind
Fremantle Doctor	Perth, Western Australia	Afternoon sea breeze that cools Perth in summer
Harmattan	Central Africa	Dry northerly wind
Kali Andhi	Indo-Gangetic Plain, Indian Subcontinent	Violent dust squalls before monsoon
Karaburan	Central Asia	Spring and winter katabatic wind
Khamsin	North Africa to Eastern Mediterranean	Hot, southeasterly wind
Loo	Plains of India and Pakistan	Hot wind
Maestro	Adriatic Sea	Cold northerly wind
Mistral	Central France and the Alps to the Mediterranean	Cold northerly wind
Monsoon	Various equatorial regions	South-westerly winds with heavy rain
Nor'easter	Northeastern USA and Atlantic Canada	A strong storm with northeast winds
Nor'wester	New Zealand's South Island	Brings rain to the West Coast and warm, dry winds to the East Coast
Pampero	Pampas, Argentina	Very strong wind
Passat	Tropical seas	Medium-strong, constant blowing wind
Rashabar	Kurdistan, Iraq	Strong "black wind"
Santa Ana Winds	Southern California, USA	Hot, dry winds
Shamal	Iraq and Persian Gulf states	Summer northwesterly wind
Simoom	Sahara, Israel, Jordan, Syria, and the Arabian Desert	Strong, dry desert wind

Sirocco	North Africa to Southern Europe	Southerly wind
Southerly Buster	Sydney, Australia	Rapidly arriving low-pressure cell
Sou'wester	Coastal areas	Strong wind from the southwest
Squamish	British Columbia, Canada	Strong, violent wind in fjords
Vendavel	Strait of Gibraltar	Westerly wind
Zonda	Argentina	Warm wind on the eastern slope of the Andes